

# What the initial-biomass back-calculation does to a respirometry thermal-performance curve

A structural analysis of  $N_0 = N_{\text{inoc}} e^{r\delta}$ , with a three-arm sensitivity re-run of the *Candida* carbon-economy results

2026-07-26

## Abstract

Closed-vial respirometry cannot count cells at the moment the oxygen fit window opens without breaking the seal that makes the measurement possible. The initial biomass must therefore be back-calculated. This report examines the back-calculation used in this project and in the published method it derives from, shows that it makes per-cell respiration an explicit decreasing function of the fitted growth rate, quantifies the consequence, and re-runs every downstream result under three alternative treatments. Three of the four load-bearing claims survive all three; one does not. A previously unused temperature trace, recorded in every raw instrument file, turns the central unmeasured quantity into something partly measurable and raises a separate question about the oxygen concentrations themselves. Nothing in the analysis pipeline was changed, and no shipped default was altered.

## Table of contents

|          |                                                                    |           |
|----------|--------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Summary</b>                                                     | <b>3</b>  |
| <b>2</b> | <b>The identity</b>                                                | <b>4</b>  |
| 2.1      | Derivation . . . . .                                               | 4         |
| 2.2      | Verification . . . . .                                             | 4         |
| 2.3      | Why this matters . . . . .                                         | 4         |
| <b>3</b> | <b>What <math>\delta</math> is</b>                                 | <b>5</b>  |
| 3.1      | The distribution of the term . . . . .                             | 5         |
| 3.2      | The evidence that it is instrumental . . . . .                     | 5         |
| 3.3      | How much of $E_R$ is the correction? . . . . .                     | 6         |
| <b>4</b> | <b>The pre-window transient</b>                                    | <b>7</b>  |
| 4.1      | Two corrections to the original reading . . . . .                  | 7         |
| 4.2      | Contamination of the fit window . . . . .                          | 8         |
| <b>5</b> | <b>The recorded internal temperature</b>                           | <b>8</b>  |
| 5.1      | What is recorded, and what is discarded . . . . .                  | 8         |
| 5.2      | What this implies for the reported oxygen concentrations . . . . . | 9         |
| 5.3      | The thermal ramp, measured rather than assumed . . . . .           | 9         |
| <b>6</b> | <b>The three arms</b>                                              | <b>10</b> |
| 6.1      | Design . . . . .                                                   | 10        |
| 6.2      | The reproducibility floor . . . . .                                | 11        |
| 6.3      | Invariance . . . . .                                               | 11        |
| 6.4      | Results . . . . .                                                  | 11        |

|           |                                                          |           |
|-----------|----------------------------------------------------------|-----------|
| 6.5       | The capacity correlations . . . . .                      | 12        |
| <b>7</b>  | <b>What survives</b>                                     | <b>13</b> |
| 7.1       | Discrepancies between the source reports . . . . .       | 13        |
| <b>8</b>  | <b>Recommendation</b>                                    | <b>13</b> |
| <b>9</b>  | <b>What to add to the assay</b>                          | <b>14</b> |
| 9.1       | Cell-free wells — at no cost in plates or runs . . . . . | 14        |
| 9.2       | One endpoint cell count per vial . . . . .               | 14        |
| 9.3       | What the two together enable . . . . .                   | 14        |
| 9.4       | What exists today . . . . .                              | 15        |
| 9.5       | One free improvement . . . . .                           | 15        |
| <b>10</b> | <b>Appendix</b>                                          | <b>16</b> |
| 10.1      | Identifiability of growth during $\delta$ . . . . .      | 16        |
| 10.2      | Method-level versus dataset-specific . . . . .           | 16        |
| 10.3      | Methods . . . . .                                        | 17        |
| 10.4      | Figures and tables cited . . . . .                       | 17        |

# 1 Summary

**What was asked.** Per-cell respiration in this project is computed as  $K/N_0$ , where  $N_0$  is the cell density at the start of the oxygen fit window, obtained by projecting the known inoculum forward across the interval  $\delta$  between filling the vial and the start of the fit:  $N_0 = N_{\text{inoc}} \cdot e^{r\delta}$ . Because  $r$  is the fitted growth rate, respiration inherits it. The question was how much of the reported between-taxon variation in respiratory temperature sensitivity is generated by that term rather than measured.

**What was found.**

1. **The coupling is structural, not incidental.** The fit-window length cancels exactly, leaving  $\log R = \log K - \log N_{\text{inoc}} - r\delta$ , verified to a residual standard deviation of  $1.3 \times 10^{-14}$ . Respiration is an explicit decreasing function of the fitted growth rate, by construction.
2. **The term is large and shaped like the growth curve.** Median  $r\delta = 0.679$ , i.e. a two-fold deflation of per-cell respiration; 95th percentile 4.0-fold, maximum 35.7-fold. It peaks at 32–34 °C — the growth optimum — and falls away at both ends. It does not add noise; it imprints an inverted growth curve on respiration.
3.  **$\delta$  is instrumental, not biological.** It is flat with temperature (58–82 min;  $R^2 = 0.013$ ), unrelated to the fitted growth rate ( $R^2 = 0.002$ ), and shows no taxon effect once *C. glabrata* is set aside ( $p = 0.097$ ). A biological lag would be roughly three times shorter near 34 °C, where growth is three times faster. It is not.
4. **Removing the term destroys the ordering the paper reports.** *C. auris* Clade IV moves from the lowest respiratory activation energy (0.291 eV) to fourth of six (0.597 eV); the rank correlation between the two orderings is +0.486 ( $p = 0.33$ ).
5. **A previously discarded temperature trace changes the picture.** Every raw instrument file records `T_internal`, a measured temperature. The plate dips 4.75 K below its settled temperature ten minutes into a 44 °C run and is still 0.40 K low at 84 minutes. The instrument’s oxygen compensation used a *constant* temperature throughout. This is new, and it is available from data already in hand.
6. **Three of four load-bearing claims survive every treatment tested.** The exception is “*Clade IV is credibly the cheapest at fever*”, which fails under the lower bound.

**What should change.** Keep the current treatment; stop describing it as either an artefact or as clean. Soften the Clade IV claim to the species-level claim that survives everywhere. Withdraw or caveat one figure panel whose correlation exists only through the coupling. Add two cheap measurements — cell-free wells in sensor positions the plate already leaves empty, and one endpoint cell count per vial — which together would replace the assumption with a measurement.

**Scope.** Findings 1–3 and 5 are **method-level**: the same construction appears in the published method (`icakin/OxygenModel`, `CUE.R:222`) and the same instrument records the same unused trace. Findings 4 and 6 are **dataset-specific**.

## 2 The identity

### 2.1 Derivation

The pipeline fits, to each oxygen series over a window of length  $T_w$ ,

$$O_2(t) = O_{2,0} + \frac{K}{r} (1 - e^{rt}),$$

with  $K$  the volumetric oxygen consumption rate at the window start ( $\text{mg L}^{-1} \text{ min}^{-1}$ ) and  $r$  the specific growth rate ( $\text{min}^{-1}$ ). It then forms

$$\begin{aligned} N_0 &= N_{\text{inoc}} e^{r\delta}, \\ C_{\text{tot}} &= \frac{K}{r} (e^{rT_w} - 1), \\ B &= N_0 \frac{e^{rT_w} - 1}{r}, \\ R_{\text{cell}} &= C_{\text{tot}}/B. \end{aligned}$$

$C_{\text{tot}}$  is the total oxygen consumed over the window;  $B$  is the integrated biomass across it. Substituting,

$$R_{\text{cell}} = \frac{\frac{K}{r} (e^{rT_w} - 1)}{N_0 \frac{e^{rT_w} - 1}{r}} = \frac{K}{N_0}.$$

**The window length cancels identically.** Nothing about how long the fit ran, or how much oxygen was consumed in total, survives into per-cell respiration. Taking logarithms and substituting  $N_0$ ,

$$\log R_{\text{cell}} = \log K - \log N_{\text{inoc}} - r\delta$$

### 2.2 Verification

Computed on all 991 retained series, the residual  $\log R - [\log K - \log N_{\text{inoc}} - r\delta]$  is constant at 30.745515174 with standard deviation  $1.3 \times 10^{-14}$  — exactly  $\log(\text{MG\_TO\_FG} \cdot \text{O2\_TO\_C\_MASS} \cdot \text{RQ} \cdot \text{MIN\_TO\_H})$ , the unit conversion. There is nothing else in the expression.

*Provenance:* `reports/tools/c2_partA.json`  $\rightarrow$  `identity_check`; source `scripts/07_oxygen_fits.R`:361, 922, 928, 935, 941.

### 2.3 Why this matters

This is a **structural property of the estimator, not a bug**. Every closed-vial per-cell rate must divide by a biomass that was not observed at the time the rate was measured, and any back-calculation will introduce *some* dependence. The question is never whether to back-calculate — it is which back-calculation is best supported, and what the chosen one does to the quantity being compared across conditions.

Here the answer is specific: because  $r$  enters  $\log R$  with a coefficient of  $-\delta$ , and because  $\delta$  is roughly constant while  $r$  varies strongly with temperature, the estimator subtracts a *scaled copy of the growth thermal-performance curve* from log respiration. Anything computed from the difference between growth and respiration sensitivities is therefore partly computing the growth curve against itself.

### 3 What $\delta$ is

#### 3.1 The distribution of the term

**Table 1:** The back-projection term across 991 retained series. The median series has its per-cell respiration deflated about two-fold by this term alone.

| quantity                 | median       | IQR             | p95    | max         |
|--------------------------|--------------|-----------------|--------|-------------|
| $\delta$ (min)           | 75.2         | 65.2 – 88.2     | 140.5  | 234.7       |
| $r$ (min <sup>-1</sup> ) | 0.0088       | 0.0058 – 0.0120 | 0.0154 | 0.0208      |
| $r\delta$                | <b>0.679</b> | 0.421 – 0.980   | 1.392  | 3.575       |
| $e^{r\delta}$            | <b>1.97</b>  | 1.52 – 2.67     | 4.02   | <b>35.7</b> |

The temperature structure is the important part:

**Table 2:** The deflation peaks at the growth optimum and falls away at both ends, tracking  $r$ . Abridged; all twelve temperatures in the source.

| T (°C)    | n  | $\delta$ median | $r\delta$ median | $e^{r\delta}$ median | $e^{r\delta}$ max |
|-----------|----|-----------------|------------------|----------------------|-------------------|
| 22        | 89 | 58.1            | 0.226            | 1.25                 | 2.27              |
| 26        | 90 | 74.2            | 0.541            | 1.72                 | 7.02              |
| 30        | 90 | 80.2            | 0.832            | 2.30                 | 7.79              |
| <b>32</b> | 90 | 78.2            | <b>0.978</b>     | <b>2.66</b>          | 14.46             |
| <b>34</b> | 89 | 76.2            | <b>0.980</b>     | <b>2.66</b>          | <b>35.68</b>      |
| 36        | 87 | 76.2            | 0.938            | 2.55                 | 16.79             |
| 40        | 78 | 79.2            | 0.743            | 2.10                 | 7.69              |
| 44        | 47 | 82.2            | 0.422            | 1.52                 | 2.80              |

*Provenance:* reports/tools/c2\_partA.json  $\rightarrow$  overall, by\_temperature.

#### 3.2 The evidence that it is instrumental

**Table 3:** Nothing biological predicts  $\delta$ .

| test                                                   | result                                                                             |
|--------------------------------------------------------|------------------------------------------------------------------------------------|
| $\delta$ vs temperature                                | slope <b>+0.489 min °C<sup>-1</sup></b> , $R^2 = 0.013$ , $p = 3.5 \times 10^{-4}$ |
| $\delta$ vs fitted $r$                                 | $R^2 = 0.002$ , $p = 0.15$                                                         |
| $\delta$ vs taxon                                      | ANOVA $F(5, 985) = 210$ , $p = 1.5 \times 10^{-152}$                               |
| $\delta$ vs taxon, <b>excluding</b> <i>C. glabrata</i> | $F(4, 818) = 1.97$ , $p = 0.097$                                                   |

Mean  $\delta$  by taxon: Clade I 69.9, Clade II 71.6, Clade III 69.8, Clade IV 73.1, *C. parapsilosis* 73.2, and *C. glabrata* **125.6** min — a residual of +44.9 min. The entire apparent taxon effect is one species with a documented signal-to-noise limit.

The decisive comparison is with what a biological lag would look like. Median  $\delta$  runs from 58.1 min at 22 °C to 82.2 min at 44 °C — a 24 min spread over a 22 °C range, on a 72 min baseline, in the *wrong direction*. Growth near 34 °C is about three times faster than at 22 °C; a lag phase would be correspondingly shorter. It is not.

#### ! Caveat, stated plainly

Every set point in this design is above the assumed 20 °C ambient, so  $|T - T_{\text{ambient}}|$  is a strictly increasing function of  $T$ : the two regressions are **numerically identical**, and this design cannot distinguish “ $\delta$  depends on distance from ambient” from “ $\delta$  depends on temperature”. The argument that the pre-window interval is instrumental therefore does **not** rest on that regression. It rests on

the transient analysis in Section 4, where amplitude and time constant can be separated, and on the recorded internal temperature in Section 5.

One point of provenance:  $\delta$  equals the automatically detected oxygen peak in only **16 of 991 series (1.6 %)**. It is the *hand-chosen fit-window start* from the interactive trimming tool. Both are “time from filling to the start of the fit”, so the argument is unaffected — but the quantity should be named correctly.

### 3.3 How much of $E_R$ is the correction?

Removing the term makes the respiratory activation energy identical to  $E_K$ , the slope of the raw volumetric consumption rate, by construction — because  $N_{\text{inoc}}$  carries no temperature dependence.

**Table 4:** Activation energies (eV) over the full 22–44 °C range. The artefact is ordered by growth rate — largest for Clade IV, which has the largest median  $r\delta$ .

| taxon                  | $E_K = E_R(\text{removed})$ | $E_R$ shipped     | artefact     | median $r\delta$ |
|------------------------|-----------------------------|-------------------|--------------|------------------|
| Clade I                | $0.625 \pm 0.030$           | $0.518 \pm 0.033$ | <b>0.107</b> | 0.675            |
| Clade II               | $0.545 \pm 0.030$           | $0.416 \pm 0.035$ | <b>0.129</b> | 0.429            |
| Clade III              | $0.592 \pm 0.020$           | $0.404 \pm 0.029$ | <b>0.189</b> | 0.733            |
| Clade IV               | $0.597 \pm 0.021$           | $0.291 \pm 0.032$ | <b>0.306</b> | 0.893            |
| <i>C. glabrata</i>     | $0.825 \pm 0.040$           | $0.694 \pm 0.080$ | 0.131        | 0.797            |
| <i>C. parapsilosis</i> | $0.547 \pm 0.043$           | $0.339 \pm 0.060$ | <b>0.207</b> | 0.656            |

**The ordering does not survive removal.**

- shipped: IV < para < III < II < I < glab
- removed: II < para < III < IV < I < glab
- Spearman between the two orderings: +0.486 ( $p = 0.33$ )
- between-taxon spread: **0.403 eV**  $\rightarrow$  **0.280 eV**

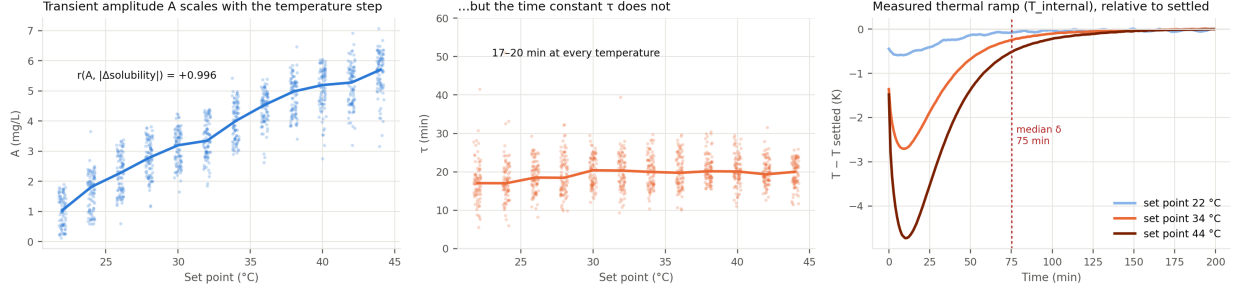
On the rising limb alone ( $T \leq 34$  °C) it is starker: shipped  $E_R$  spans 0.054–0.358 eV and Clade IV’s is  $0.054 \pm 0.057$ , indistinguishable from zero; with the term removed all six lie between 0.55 and 0.86 eV.

These simple slopes track the project’s hierarchical Bayesian  $E_R$  to within **0.024 eV** for every taxon (Clade I 0.5176 vs 0.5193; Clade IV 0.2906 vs 0.3006), so the arithmetic here is a faithful stand-in for the fitted model.

*Provenance:* `reports/tools/c2_partA.json`  $\rightarrow$  `slopes_full`, `slopes_rising`; comparison against `runs/C1_reproduction/tables/fig_values.csv`.

## 4 The pre-window transient

The pre-window transient is instrumental, and the thermal ramp is measurable



**Figure 1:** Left: the amplitude of the pre-window oxygen transient scales with the temperature step. Centre: its time constant does not — it is 17–20 min at every set point. Right: the measured internal temperature, relative to its settled value, showing the dip-and-recovery discussed in Section 5. The dashed line marks the median  $\delta$ .

Every series shows a first-order approach to equilibrium before the oxygen peak. Fitting  $O_2(t) = O_{2,eq} - Ae^{-t/\tau}$  to the pre-peak segment of all 1080 series converges for **1074** (six do not rise), with median  $R^2 = 0.9988$  (IQR 0.9980–0.9994) and median RMSE 0.027 mg L<sup>-1</sup>.

**Table 5:** Median [IQR] of the fitted transient. Abridged; all twelve temperatures in the source.

| T (°C) | A (mg L <sup>-1</sup> ) | $\tau$ (min)      | $O_{2,eq}$ | settled at the peak |
|--------|-------------------------|-------------------|------------|---------------------|
| 22     | 1.04 [0.54, 1.43]       | 17.0 [14.1, 19.3] | 10.74      | 98.0 %              |
| 28     | 2.78 [2.25, 3.27]       | 18.4 [15.2, 22.8] | 10.45      | 98.3 %              |
| 34     | 4.01 [3.39, 4.47]       | 20.0 [16.7, 23.0] | 9.02       | 97.7 %              |
| 38     | 4.98 [4.26, 5.27]       | 20.1 [18.3, 22.5] | 8.96       | 97.6 %              |
| 44     | 5.70 [5.21, 6.08]       | 20.0 [17.4, 22.5] | 8.97       | 98.1 %              |

**It is physico-chemical.** The amplitude scales with the temperature step ( $A \sim |T - 20|$ : slope +0.2035 mg L<sup>-1</sup> °C<sup>-1</sup>,  $R^2 = 0.822$ , Spearman  $\rho = +0.910$ ), and against the Benson–Krause oxygen solubility curve,

$$r(A, |\text{sol}(T) - \text{sol}(20)|) = +0.996, \quad p = 9.5 \times 10^{-12},$$

with the ratio settling at about 1.9–2.0 for  $T \geq 32$  °C. The time constant shows **no taxon effect** once *C. glabrata* is excluded (ANOVA on per-temperature residuals,  $F(4, 889) = 1.99$ ,  $p = 0.094$ ); *C. glabrata* is +2.8 min on a 19 min baseline.

### 4.1 Two corrections to the original reading

**i** does not scale with distance from ambient

The time constant is 17.0–20.4 min at **every** set point (slope +0.108 min °C<sup>-1</sup>,  $R^2 = 0.027$ ). What lengthens at high temperature is the time to the *peak*, and only because a larger rise takes longer to be overtaken by the biological drawdown. Figures of 37, 51 and 70 min quoted in earlier framing of this problem are **peak times, not time constants**. What scales is the amplitude.

**i** The sign is inverted

Warming should *lower* dissolved oxygen; the record *rises*. The fitted asymptote sits **1.20–1.48× above air saturation**, with the excess growing with temperature, while tracking the solubility curve in shape (Pearson  $r = +0.892$  across the twelve set points, slope  $-0.091$  vs  $-0.121 \text{ mg L}^{-1} \text{ }^{\circ}\text{C}^{-1}$ ). This is not gas re-equilibration. Section 5 identifies what it is.

## 4.2 Contamination of the fit window

Because the transient is not perfectly complete when the window opens, some residual abiotic drift sits inside it. Its sign matters: the transient *rises*, so it masks part of the biological drawdown and makes  $K$  **too small**.

Measured by extrapolating each fitted transient forward, subtracting it from the observed trace over the actual fit window, and refitting the project’s own oxygen model with its own bounds — a procedure first validated by reproducing the shipped  $K$  from the raw window (median relative difference  $4.9 \times 10^{-3}$ ; 66 % within 1 %).

**Table 6:** A sensitivity band, not a point estimate — there are no cell-free wells anywhere in this design, so the in-window transient is extrapolated, never observed.

| T ( $^{\circ}\text{C}$ ) | drift ( $\text{mg L}^{-1}$ ) | as % of drawdown | $K$ under-read: $\tau_{q25} \rightarrow$<br><b>fitted</b> $\tau \rightarrow \tau_{q75}$ |
|--------------------------|------------------------------|------------------|-----------------------------------------------------------------------------------------|
| 22                       | 0.016                        | 0.52 %           | 0.21 % $\rightarrow$ <b>0.54</b> % $\rightarrow$<br>0.99 %                              |
| 28                       | 0.047                        | 1.02 %           | 0.76 % $\rightarrow$ <b>2.16</b> % $\rightarrow$<br>5.34 %                              |
| 34                       | 0.075                        | 1.83 %           | 2.42 % $\rightarrow$ <b>5.02</b> % $\rightarrow$<br>10.46 %                             |
| 36                       | 0.093                        | 2.40 %           | 3.02 % $\rightarrow$ <b>6.41</b> % $\rightarrow$<br>13.48 %                             |
| 40                       | 0.106                        | 2.46 %           | 3.65 % $\rightarrow$ <b>6.63</b> % $\rightarrow$<br>11.09 %                             |
| 44                       | 0.097                        | 2.27 %           | 2.14 % $\rightarrow$ <b>4.01</b> % $\rightarrow$<br>7.35 %                              |

It **does** grow with temperature ( $+0.288$  percentage points  $^{\circ}\text{C}^{-1}$ ,  $p = 1.3 \times 10^{-23}$ ). **But it cannot explain the monotonic rise of respiration to 44  $^{\circ}\text{C}$ , because correcting it makes the rise steeper.**  $E_K$  after correction at the fitted  $\tau$ : Clade I  $0.621 \rightarrow 0.656$ , Clade IV  $0.569 \rightarrow 0.591$ , *C. parapsilosis*  $0.540 \rightarrow 0.574$  — and the entire  $\tau$ -IQR band preserves that sign.

*Provenance:* reports/tools/c2\_partB.json, c2\_partB\_per\_series.csv.

## 5 The recorded internal temperature

**!** This is a new finding, and it needs no new experiment

Everything in this section is a recomputation from data already in the repository. No sample was re-run, no instrument re-read.

### 5.1 What is recorded, and what is discarded

Each raw instrument export carries 32 columns. The last four are  $T_m$  [ $^{\circ}\text{C}$ ],  $p$  [mbar], Salinity [g/1000g] and  $T_{\text{internal}}$  [ $^{\circ}\text{C}$ ]. The conversion script (scripts/01\_convert\_xlsx.R) keeps only Time, the plate temperature label, and the 24 well traces.  **$T_{\text{internal}}$  and  $T_m$  are discarded**, and with them the only direct evidence of what the sample temperature actually did.

Recovering them from all 72 plates (99,120 rows) shows two distinct things.

**Tm is a constant.** It equals the file-name set point in **72 of 72 plates** and does not vary within a run. It is the value typed into the instrument software. With **Temperature comp. : ON** in the calibration header, it is what the optode’s temperature compensation used for the whole run. The same header records **02\_T0 = 02\_T2nd = 23.2 °C**: the sensor was two-point calibrated at 23.2 °C.

**T\_internal is a measurement**, and it is not flat.

**Table 7:** Measured internal temperature, mean of the six plates at each set point. Abridged; all twelve in the source.

| set point | asymptote    | dip below<br>asymptote | at $t$ (min) | still low at 84 min | set point –<br>asymptote |
|-----------|--------------|------------------------|--------------|---------------------|--------------------------|
| 22        | 22.66        | 0.57 K                 | 6.0          | 0.02 K              | −0.66                    |
| 30        | 29.44        | 2.08 K                 | 11.0         | 0.22 K              | +0.56                    |
| 34        | 33.74        | 2.73 K                 | 9.0          | 0.21 K              | +0.26                    |
| 40        | 40.11        | 4.02 K                 | 10.0         | 0.33 K              | −0.11                    |
| 44        | <b>43.55</b> | <b>4.75 K</b>          | 10.0         | <b>0.40 K</b>       | +0.45                    |

Per-plate values differ slightly from the six-plate mean. The Clade I plate at 44 °C — the one quoted in earlier framing of this work — dips **4.79 K** at 10.5 min, is 0.31 K low at 84 min, and asymptotes at **43.47 °C**. Both are correct; they are different aggregations, and this report uses the six-plate mean unless stated.

Two further observations. The plate **does not start at ambient**: at a 44 °C set point it begins 1.6 K below its own asymptote, dips as the load equilibrates, and recovers over roughly an hour. And the asymptote is not the set point — the mean absolute gap across the twelve set points is 0.35 K, with the 44 °C plates settling 0.45 K short.

## 5.2 What this implies for the reported oxygen concentrations

The consequence is not confined to the thermal-ramp correction. Oxygen concentration in  $\text{mg L}^{-1}$  is derived from the optode’s phase reading using a temperature. The instrument used a **constant** temperature — the set point — while the sample was measurably up to **4.75 K cooler**, over **exactly the pre-window interval** on which  $\delta$ , the transient and  $K$ ’s starting slope all depend.

That is a coherent single explanation for three separate observations in Section 4: why a transient exists at all, why its amplitude scales so tightly with the temperature step ( $r = +0.996$  against the solubility change), and why the fitted asymptote sits above air saturation by a margin that grows with temperature.

 Open, and deliberately left open

**This correction has not been applied.** Applying it would mean recomputing oxygen concentrations from phase and the measured temperature, which is an analysis change and outside the scope of this report. Until it is done, the fit-window contamination bound in Table 6 rests on a transient that is itself partly a compensation artefact, and the question of how much of  $K$ ’s temperature dependence is instrumental **remains open**. The direction of the effect on  $E_K$  (Section 4) is robust; its magnitude is not settled.

## 5.3 The thermal ramp, measured rather than assumed

With the measured trajectory in hand, the ramp-aware initial biomass is

$$N_0^{\text{ramp}} = N_{\text{inoc}} \exp \left( \int_0^\delta r(T(t)) dt \right),$$

with  $T(t)$  the measured trace and  $r(T)$  each isolate’s own thermal-performance curve, scaled through that series’ own fitted  $r$  so that only the *shape* of the curve is used.

**Table 8:** The ratio is not monotone, and the growth optimum is why.

| T (°C)    | mean $T$ over $[0, \delta]$ | $N_0^{\text{ramp}}/N_0^{\text{current}}$ |
|-----------|-----------------------------|------------------------------------------|
| 22        | 22.37                       | 0.9920 [0.9898, 0.9943]                  |
| 30        | 28.37                       | 0.9208 [0.9143, 0.9321]                  |
| <b>32</b> | 30.00                       | <b>0.9150</b> [0.8951, 0.9315]           |
| 36        | 34.27                       | 0.9802 [0.9585, 1.0232]                  |
| 40        | 38.22                       | 1.1394 [1.0918, 1.2022]                  |
| 44        | 41.44                       | <b>1.1996</b> [1.1395, 1.2724]           |

Below about 36 °C the ramp sits below the set point, the cells grew slower than  $r$ , and the current  $N_0$  **over-corrects**. Above it the ramp passes *through* the optimum, the cells grew *faster* than at the supra-optimal set point, and the current  $N_0$  **under-corrects**. The net effect on  $E_R$  is  $-0.009$  to  $-0.102$  eV depending on taxon.

An earlier version of this correction, which assumed a start at 20 °C ambient and the oxygen signal’s 19 min time constant, **overstated the ramp roughly two-fold** (0.841 vs 0.941 at 34 °C). That version is retained for comparison as `reports/tools/c2_ramp_n0_factors_modelled.csv`.

**i** Why this is a bound and not an estimator

The growth thermal-performance curve is invariant to  $N_0$  —  $r$  comes from the shape of the oxygen curve, and  $N_0$  enters only through  $R_{\text{cell}} = K/N_0$  — so there is no feedback through the growth side. The real coupling is the **342 of 782 series (43.7 %)** still more than 0.25 K below settled when their fit window opens, where the fitted  $r$  is itself a ramp average. One iteration was performed; the second-iteration change in the ratio is median 0.0048, 95th percentile 0.065, maximum 0.360.  $T_{\text{internal}}$  is also the reader’s internal sensor, not a probe in the liquid. It remains a *proxy* for vial temperature — but a **measured** proxy, which the previous assumption was not.

## 6 The three arms

### 6.1 Design

**Table 9:** Each arm is a complete re-run of stages 07 → 08 → 09 → 10 → 12 → 13 with a full Bayesian refit.

|              | treatment                                                                           | wall time |
|--------------|-------------------------------------------------------------------------------------|-----------|
| <b>ARM 1</b> | $N_0 = N_{\text{inoc}} e^{r\delta}$ — the shipped default                           | 10 m 33 s |
| <b>ARM 2</b> | $N_0 = N_{\text{inoc}} e^{r\delta} f_{\text{ramp}}$ — ramp-aware                    | 10 m 23 s |
| <b>ARM 3</b> | $N_0 = N_{\text{inoc}}$ — <b>a lower bound on <math>N_0</math>, not a candidate</b> | 11 m 07 s |

Everything but the  $N_0$  treatment was held fixed: the committed trim windows, exclusions, inoculum and cell-carbon constants; every model formula, prior and the sampler seed; and the genome-scale-model capacity values. The stages that precede  $N_0$  are shared byte-for-byte across arms.

**i** Why ARM 3 is reported despite being known to be wrong

$\delta$  is a *detection* delay, so asserting zero growth across it is demonstrably false, and ARM 3 is not the preferred estimator. It is retained as a **diagnostic**, because it is the only arm that removes the  $r$

dependence entirely. Where a result changes between ARM 1 and ARM 3, that change isolates the mechanism: it shows the result is carried by the coupling rather than by the measurement. That is informative even though the arm itself is a bound.

## 6.2 The reproducibility floor

Before attributing any difference to  $N_0$ , the noise floor was measured by re-running the shipped treatment and comparing against the independent earlier reproduction: **35 of 36 comparable tables identical to better than  $10^{-10}$** , the single exception being one curve file at 0.165 absolute / 2.11 % relative — inside the previously measured unseeded floor of 0.205 / 2.21 %. Four previously unseeded subsampling stages were given one recorded seed, applied identically across arms; this **moved no point estimate**.

Every difference reported below is therefore against a floor of **exactly zero**, except the Figure 2a curve band, on which no claim rests.

## 6.3 Invariance

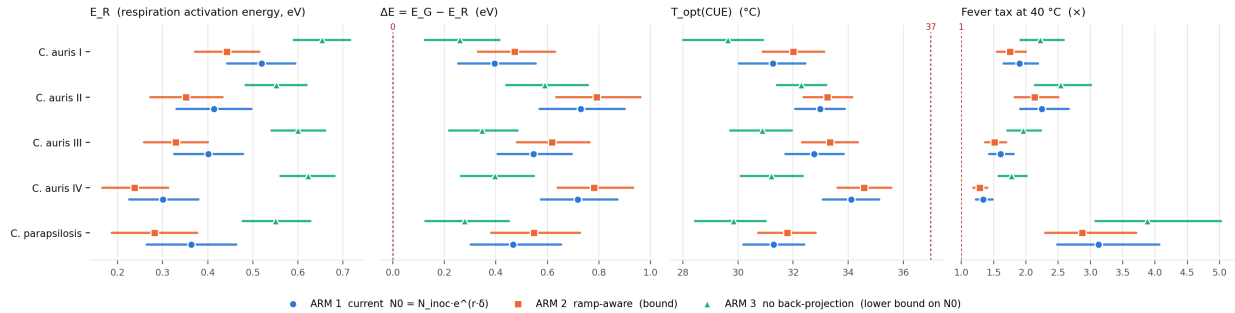
$N_0$  enters only respiration, so growth must not move. Maximum absolute difference between every pair of arms, for  $r$ ,  $K$ , absolute growth, mass-specific growth, the growth posterior summary, the per-isolate growth curve coefficients, the Arrhenius growth coefficients,  $E_G$  and the growth optimum:

$$0.000 \times 10^0 \text{ in every case.}$$

The switch does not leak.

## 6.4 Results

**N0 sensitivity: the three treatments, with 95% credible intervals**  
Growth-only quantities are bit-identical across arms; only respiration moves. Every value is tabulated in reports/NQ\_SENSITIVITY.md.



**Figure 2:** The three treatments across the four load-bearing quantities, all taxa, with 95 % credible intervals. Growth-side quantities are bit-identical across arms; only respiration moves. Each arm carries a distinct marker shape as well as a colour, so identity never rests on colour alone.

**Table 10:**  $E_R$  (eV), posterior median [95 % credible interval].  $E_G$  is identical in all three arms: Clade I 0.914, Clade II 1.143, Clade III 0.947, Clade IV 1.019, *C. parapsilosis* 0.830.

| taxon                  | ARM 2 ramp           | ARM 1 current               | ARM 3 lower bound    |
|------------------------|----------------------|-----------------------------|----------------------|
| Clade I                | 0.442 [0.372, 0.514] | <b>0.519 [0.443, 0.594]</b> | 0.654 [0.591, 0.716] |
| Clade II               | 0.352 [0.273, 0.433] | <b>0.414 [0.331, 0.497]</b> | 0.553 [0.484, 0.620] |
| Clade III              | 0.329 [0.259, 0.400] | <b>0.401 [0.326, 0.479]</b> | 0.600 [0.541, 0.660] |
| Clade IV               | 0.238 [0.167, 0.312] | <b>0.301 [0.226, 0.380]</b> | 0.622 [0.561, 0.682] |
| <i>C. parapsilosis</i> | 0.283 [0.188, 0.377] | <b>0.363 [0.264, 0.463]</b> | 0.551 [0.478, 0.628] |

**Table 11:**  $\Delta E = E_G - E_R$  (eV). Excludes zero in every taxon in every arm.

| taxon                  | ARM 2 ramp           | ARM 1 current               | ARM 3 lower bound    |
|------------------------|----------------------|-----------------------------|----------------------|
| Clade I                | 0.472 [0.331, 0.631] | <b>0.395</b> [0.254, 0.556] | 0.261 [0.125, 0.414] |
| Clade II               | 0.791 [0.634, 0.962] | <b>0.730</b> [0.570, 0.900] | 0.590 [0.441, 0.758] |
| Clade III              | 0.619 [0.482, 0.763] | <b>0.547</b> [0.407, 0.695] | 0.347 [0.219, 0.485] |
| Clade IV               | 0.781 [0.641, 0.934] | <b>0.718</b> [0.576, 0.872] | 0.397 [0.265, 0.548] |
| <i>C. parapsilosis</i> | 0.547 [0.382, 0.727] | <b>0.468</b> [0.303, 0.653] | 0.279 [0.127, 0.451] |

**Table 12:** Carbon-economy optimum  $T_{\text{opt}}$  (CUE), °C.  $P(T_{\text{opt}} < 37 \text{ °C}) = 1.0000$  for every taxon in every arm.

| taxon                  | ARM 2 ramp           | ARM 1 current               | ARM 3 lower bound    |
|------------------------|----------------------|-----------------------------|----------------------|
| Clade I                | 32.01 [30.90, 33.14] | <b>31.27</b> [30.03, 32.45] | 29.63 [28.01, 30.90] |
| Clade II               | 33.25 [32.38, 34.15] | <b>32.98</b> [32.08, 33.88] | 32.30 [31.42, 33.20] |
| Clade III              | 33.35 [32.32, 34.35] | <b>32.77</b> [31.72, 33.83] | 30.88 [29.72, 31.96] |
| Clade IV               | 34.58 [33.61, 35.57] | <b>34.11</b> [33.08, 35.13] | 31.21 [30.10, 32.35] |
| <i>C. parapsilosis</i> | 31.78 [30.73, 32.81] | <b>31.30</b> [30.20, 32.40] | 29.84 [28.44, 31.01] |

**Table 13:** The ten pairwise fever-cost contrasts. marks a 95 % credible interval excluding 1. **Resolved: 8/10, 8/10, 7/10.** The one that flips is Clade III vs Clade IV.

| a                | b               | ARM 1 current               | ARM 2 ramp                  | ARM 3 lower bound           |
|------------------|-----------------|-----------------------------|-----------------------------|-----------------------------|
| Clade IV         | para            | 0.430 [0.326, 0.551]        | 0.447 [0.341, 0.565]        | 0.459 [0.346, 0.595]        |
| Clade III        | para            | 0.514 [0.386, 0.668]        | 0.526 [0.398, 0.673]        | 0.504 [0.376, 0.657]        |
| Clade I          | para            | 0.607 [0.452, 0.797]        | 0.611 [0.460, 0.789]        | 0.573 [0.426, 0.756]        |
| Clade II         | para            | 0.718 [0.523, 0.956]        | 0.741 [0.547, 0.973]        | 0.656 [0.479, 0.871]        |
| Clade I          | Clade II        | 0.846 [0.683, 1.043]        | 0.823 [0.672, 1.009]        | 0.873 [0.698, 1.091]        |
| Clade I          | Clade III       | 1.181 [0.983, 1.414]        | 1.161 [0.980, 1.369]        | 1.134 [0.927, 1.386]        |
| <b>Clade III</b> | <b>Clade IV</b> | <b>1.198</b> [1.031, 1.394] | <b>1.176</b> [1.029, 1.352] | <b>1.100</b> [0.920, 1.319] |
| Clade II         | Clade III       | 1.395 [1.143, 1.711]        | 1.410 [1.172, 1.704]        | 1.298 [1.050, 1.615]        |
| Clade I          | Clade IV        | 1.416 [1.200, 1.666]        | 1.365 [1.172, 1.592]        | 1.247 [1.028, 1.519]        |
| Clade II         | Clade IV        | 1.674 [1.390, 2.028]        | 1.662 [1.383, 1.987]        | 1.429 [1.162, 1.760]        |

Spearman  $\rho$  between environmental optimum and fever cost is  $-1.00 [-1.00, -0.70]$ ,  $P(\rho < 0) = 1.000$ , in all three arms.

*Provenance:* reports/tools/c2\_arms.json; per-arm tables under runs/C2\_arm\_{current,ramp,nobp}/tables/.

## 6.5 The capacity correlations

**Table 14:** Capacity is held fixed across arms, so only the economics axis moves.

| arm                  | $r(\text{capacity, fever tax at } 40 \text{ °C})$ | $r(\text{capacity, per-cell respiration})$ |
|----------------------|---------------------------------------------------|--------------------------------------------|
| <b>ARM 1 current</b> | $-0.847$ ( $p = 5.0 \times 10^{-4}$ )             | $-0.620$ ( $p = 3.2 \times 10^{-2}$ )      |
| ARM 2 ramp           | $-0.874$ ( $p = 2.1 \times 10^{-4}$ )             | $-0.603$ ( $p = 3.8 \times 10^{-2}$ )      |
| ARM 3 lower bound    | $-0.819$ ( $p = 1.1 \times 10^{-3}$ )             | $-0.196$ ( $p = 0.54$ )                    |

The first correlation is stable. The second **collapses**. Under ARM 3, per-cell respiration is  $K/N_{\text{inoc}}$  and contains no  $r$ ; capacity is fitted to growth. The correlation therefore exists only while respiration carries the  $e^{\tau\delta}$  term. This is the clearest instance of ARM 3 working as a diagnostic rather than as an estimate.

## 7 What survives

**Table 15:** The four load-bearing claims under all three treatments.

| claim                                                                         | ARM 1 current                                         | ARM 2 ramp                                  | ARM 3 lower bound                                                                                                                |
|-------------------------------------------------------------------------------|-------------------------------------------------------|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| growth is \$2× more temperature-sensitive than respiration                    | <b>SURVIVES</b> — $E_G/E_R$<br>1.76–3.39, median 2.36 | <b>SURVIVES</b> — 2.07–4.28,<br>median 2.94 | <b>SURVIVES as direction, WEAKENS as “2×”</b> —<br>1.40–2.07, median 1.58; $\Delta E$<br>still excludes zero in all five<br>taxa |
| Clade IV is credibly the cheapest at fever                                    | <b>SURVIVES</b> — III vs IV<br>1.198 [1.031, 1.394]   | <b>SURVIVES</b> — 1.176 [1.029,<br>1.352]   | <b>FAILS</b> — 1.100 [0.920,<br>1.319]                                                                                           |
| all four <i>C. auris</i> clades pay credibly less than <i>C. parapsilosis</i> | <b>SURVIVES</b> (4/4)                                 | <b>SURVIVES</b> (4/4)                       | <b>SURVIVES</b> (4/4)                                                                                                            |
| every economic optimum lies below 37 °C                                       | <b>SURVIVES</b> ( $P = 1.0000$ ,<br>×5)               | <b>SURVIVES</b> ( $P = 1.0000$ ,<br>×5)     | <b>SURVIVES</b> ( $P = 1.0000$ ,<br>×5)                                                                                          |

**Three of four survive everywhere, and one strengthens.** The headline result — that the carbon-economy optimum of every taxon lies below human body temperature — is *unaffected* by the treatment, with posterior probability 1.0000 in all fifteen taxon-arm combinations. That is a stronger statement than the paper currently makes, and it should be made.

The species-level claim is equally robust: all four *C. auris* clades pay credibly less than *C. parapsilosis* at fever in every arm, twelve of twelve contrasts resolved.

What does not survive is the finest distinction in the set — the ordering *within* the *C. auris* clades, which rests on a single contrast that the treatment can overturn.

### 7.1 Discrepancies between the source reports

None material. One clarification: the internal-temperature dip figures differ depending on aggregation. Values of 4.76 K (44 °C), 2.92 K (34 °C) and 0.75 K (22 °C), with a 43.44 °C asymptote, are the **Clade I plate alone**; the six-plate means used here are 4.75 K, 2.73 K and 0.57 K with a 43.55 °C asymptote. Both are correct. This report states which aggregation is in use.

## 8 Recommendation

**Build on ARM 1, and describe it accurately.** It is a directionally correct correction with a measurable, quantified bias. Two descriptions of it in the codebase — one calling the term “the artefact”, the other instructing that it never be changed — were both unsupported by any analysis and have been replaced with pointers to this work.

ARM 1 and ARM 2 bracket the defensible range and agree on every claim. ARM 2 says the current correction is a little too strong below 36 °C and a little too weak above it: real, quantified, and not claim-changing.

**Two things should change in the manuscript now, without waiting for new data.**

1. **Soften “Clade IV is credibly the cheapest at fever.”** It rests entirely on one contrast that fails under the lower bound. The species-level claim survives everywhere and is the safe headline.
2. **Withdraw or caveat the capacity-versus-respiration panel.** Its correlation exists only through the  $e^{r\delta}$  coupling. The capacity-versus-fever-tax panel is stable across arms and does not have this problem.

**One thing should be strengthened.** “Every economic optimum lies below body temperature” is robust to a treatment that moves  $E_R$  by up to 0.32 eV. Say so.

**What remains unresolved.** The estimand is **not identified** by any arm (Section 10.1). The magnitude of growth during  $\delta$  is *bounded* by this analysis, not measured. Separately, the temperature-compensation mismatch in Section 5 has **not** been corrected, so the fit-window contamination question is open.

## 9 What to add to the assay

### 9.1 Cell-free wells — at no cost in plates or runs

Each 24-well plate currently uses rows A–C and columns 1–5. **Row D and column 6 are “No Sensor”** — nine unused sensor positions per plate.

- **What:** wells with medium only, no inoculum, otherwise identical — same volume, same seal, same optode, same run.
- **How many:** two per plate, in positions that are already empty.
- **When:** every run.
- **What it buys:** the abiotic oxygen trajectory observed *directly through the entire fit window*, so  $K$  can be corrected rather than bounded, and Table 6 becomes a measurement instead of a sensitivity band. Combined with  $T_{\text{internal}}$ , it would also settle whether the compensation mismatch in Section 5 materially distorts the reported concentrations.

### 9.2 One endpoint cell count per vial

- **What:** a cell count from each vial immediately after the seal is broken at the end of the run.
- **How many:** one per vial — all fifteen per plate, since the value is used paired with that vial’s own  $r$  and window.
- **When:** at the end of every run. **The sealed measurement is never disturbed.**

### 9.3 What the two together enable

They make the alternative back-calculation available:

**Table 16:** The two back-calculations.

|                                   | FORWARD (current)                                                                                                                                                                  | BACKWARD (enabled)                                                                            |
|-----------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| formula<br>assumes                | $N_0 = N_{\text{inoc}} e^{r\delta}$<br>growth at rate $r$ through an interval<br>the oxygen record cannot see, and<br>during which the sample was<br>measurably not at temperature | $N_0 = N_{\text{end}} e^{-rT_w}$<br>exponential growth over the fit window<br>only            |
| extra assumption beyond the model | <b>yes</b> — extrapolates outside the fitted<br>window                                                                                                                             | <b>none</b> — the oxygen model already<br>assumes exactly this, over exactly this<br>interval |

Running **both** yields the missing quantity empirically. The ratio

$$\frac{N_{\text{end}} e^{-rT_w}}{N_{\text{inoc}} e^{r\delta}}$$

*measures* how much growth actually occurred during equilibration — the quantity this entire report has had to bound rather than observe. One experiment would settle it, and would settle it for the published method as well.

**i** Not proposed

Measuring cell density in the vial at the window start. It cannot be done without breaking the seal and destroying the closed-system measurement. The back-calculation is a considered design choice; the only question is which one.

## 9.4 What exists today

**No endpoint biomass exists anywhere in this repository.** Every column of all 72 raw exports was checked; the well traces, `Tm`, `p`, `Salinity`, `T_internal` and `Error` are all that is recorded. There is no optical density, no cell count, no biomass. The per-isolate inoculum table is the *start*, which the forward route already uses. The backward route cannot be run retrospectively here.

The published method shares this construction and is reported to have flow-cytometry data for its taxa. If those are *endpoint* counts per vial, that dataset could support this comparison retrospectively. **Flagged; not obtained, and no claim is made about what it would show.**

## 9.5 One free improvement

`scripts/01_convert_xlsx.R` should stop discarding `T_internal` and `Tm`. `T_internal` is what turned the thermal ramp from an assumption into a measurement, and it costs one column in the tidy output.

## 10 Appendix

### 10.1 Identifiability of growth during $\delta$

Let  $\varphi$  be the **true** log-growth accumulated over the pre-window interval,

$$\varphi = \int_0^\delta r_{\text{true}}(t) dt,$$

so that the true initial biomass is  $N_0^{\text{true}} = N_{\text{inoc}} e^\varphi$ . Let  $\psi$  be whatever the analysis **assumes** in its place. The three arms are three choices of  $\psi$ :

$$\psi_{\text{ARM 1}} = r\delta, \quad \psi_{\text{ARM 2}} = \int_0^\delta r(T(t)) dt, \quad \psi_{\text{ARM 3}} = 0.$$

Then  $\widehat{R} = K/(N_{\text{inoc}} e^\psi)$  and

$$\log \widehat{R} = \log R_{\text{true}} + (\varphi - \psi).$$

Two consequences follow.

**First, only the difference  $\varphi - \psi$  matters, and only its temperature dependence.** Writing  $x$  for the Boltzmann variable, the bias in the respiratory activation energy is

$$\widehat{E_R} - E_R^{\text{true}} = -\frac{\partial}{\partial x} (\varphi - \psi).$$

An estimator is unbiased for  $E_R$  **if and only if  $\varphi - \psi$  is constant in temperature** — not if  $\psi$  is close to  $\varphi$  in magnitude. ARM 1 fails this because  $\psi = r\delta$  tracks the growth curve while  $\varphi$  tracks the growth curve *along the actual thermal trajectory*; the two differ by an amount that itself depends on temperature (Table 8).

**Second, and more fundamentally:  $\varphi$  is not identified by the oxygen record.** The likelihood is built entirely from the window  $[\delta, \delta + T_w]$ .  $\varphi$  enters only as a multiplicative constant on  $N_0$ , and  $N_0$  is not a parameter of the fitted oxygen curve at all — it is applied afterwards. The profile likelihood in  $\varphi$  is therefore **exactly flat**: no value of  $\varphi$  fits the data better than any other.

This is worth stating plainly, because it determines what the three arms are. **They are not three estimates. They are three priors on an unidentified parameter.** No amount of additional modelling of the existing data can distinguish them, which is why the recommendation in §9 is a measurement rather than a method.

An endpoint count identifies  $\varphi$  directly and per series:

$$N_{\text{end}} = N_{\text{inoc}} e^\varphi e^{rT_w} \implies \varphi = \log \frac{N_{\text{end}}}{N_{\text{inoc}}} - rT_w.$$

One number per vial, taken after the experiment is over, converts a flat likelihood into a point estimate.

### 10.2 Method-level versus dataset-specific

**Table 17:** What generalises beyond this dataset.

| finding                                                                               | scope                                                                                             |
|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| $R_{\text{cell}} = K/N_0$ exactly; $\log R = \log K - \log N_{\text{inoc}} - r\delta$ | <b>method-level</b> — the construction is shared with <code>icakin/OxygenModel</code> , CUE.R:222 |
| $\varphi$ is unidentified by the oxygen record                                        | <b>method-level</b> — structural                                                                  |
| $\delta$ is a detection delay, not a lag                                              | <b>method-level</b> , subject to the same instrument                                              |
| the pre-window transient and its solubility scaling                                   | <b>method-level</b> , subject to the same instrument                                              |
| $T_{\text{internal}}$ recorded and discarded; compensation at a constant temperature  | <b>method-level</b> — same instrument, same converter                                             |
| magnitude of $r\delta$ (median 0.679)                                                 | dataset-specific                                                                                  |
| the $E_R$ ordering and its collapse                                                   | dataset-specific                                                                                  |
| the four claims and which survive                                                     | dataset-specific                                                                                  |

### 10.3 Methods

Analysis stages 02–06 and the trimming decisions were unchanged and shared across arms. Each arm re-ran stages  $07 \rightarrow 08 \rightarrow 09 \rightarrow 10 \rightarrow 12 \rightarrow 13$  with a complete Bayesian refit (4 chains  $\times$  4000 iterations, warmup 1000, fixed seed; maximum  $\widehat{R} = 1.00297$ , minimum bulk effective sample size 2601, no divergent transitions, identically in all three arms). Transient fits used bounded nonlinear least squares on the pre-peak segment of each series. Oxygen solubility used the Benson–Krause formulation for fresh water at 1 atm. Activation energies quoted outside the Bayesian tables are ordinary least-squares Boltzmann–Arrhenius slopes, which reproduce the hierarchical estimates to within 0.024 eV.

**No analysis was re-run for this report**, and no result was altered. The figures and every number are taken from the artefacts listed below.

### 10.4 Figures and tables cited

**Table 18:** Every figure in `reports/figures_NO/` is accounted for; the nine per-arm copies are retained as evidence but not reproduced here for the reasons given.

| item                        | path                                                                                        | used                                                                                             |
|-----------------------------|---------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|
| Figure 1                    | <code>reports/figures_NO/c2_transient_diagnostics.png</code>                                | yes                                                                                              |
| Figure 2                    | <code>reports/figures_NO/c2_arm_comparison.png</code>                                       | yes                                                                                              |
| Fig 1 (decoupling), per arm | <code>reports/figures_NO/FIG1_decoupling_{current,ramp,nobp}.png</code>                     | no; the three panels are bit-identical across arms, so the three copies carry no arm information |
| Fig 2 (the bill), per arm   | <code>reports/figures_NO/FIG2_the_bill_{current,ramp,nobp}.png</code>                       | no; the figure is approximate content is Table 12 and Table 13, which are exact                  |
| Fig 4 (model), per arm      | <code>reports/figures_NO/FIG_MODEL_{current,ramp,nobp}.png</code>                           | no; the figure is approximate content is Table 14                                                |
| PART A statistics           | <code>reports/tools/c2_partA.json</code>                                                    | yes                                                                                              |
| transient fits              | <code>reports/tools/c2_partB.json</code> ,<br><code>c2_partB_per_series.csv</code>          | yes                                                                                              |
| measured temperature traces | <code>reports/tools/c2_Tinternal.csv</code> ,<br><code>c2_Tinternal_dip.json</code>         | yes                                                                                              |
| measured ramp factors       | <code>reports/tools/c2_partB5_measured.json</code> ,<br><code>c2_ramp_n0_factors.csv</code> | yes                                                                                              |
| arm comparison              | <code>reports/tools/c2_arms.json</code>                                                     | yes                                                                                              |
| per-arm outputs             | <code>runs/C2_arm_{current,ramp,nobp}/tables/</code>                                        | yes                                                                                              |
| reproduction baseline       | <code>runs/C1_reproduction/tables/</code> ,<br><code>reports/REPRODUCTION.md</code>         | yes                                                                                              |
| full sensitivity record     | <code>reports/NO_SENSITIVITY.md</code>                                                      | yes                                                                                              |
| environment                 | <code>env/versions.json</code>                                                              | yes                                                                                              |